

Assignment 2: UAV Vertical Motion Simulation and PID Control

April 30, 2026

1 Objective

In this assignment, you will simulate the vertical motion of a UAV and analyze the effect of open-loop thrust and feedback control.

You will study:

1. Hovering condition
2. Open-loop drift
3. Proportional control
4. Proportional-derivative (PD) control
5. Proportional-integral-derivative (PID) control

2 Step 1: Hovering Condition

Task

Set:

- $T = mg$
- $K_p = 0, K_i = 0, K_d = 0$

Run the simulation.

The equation of motion is:

$$m\ddot{z} = T - mg \tag{1}$$

Questions

1. What happens to the altitude?
2. Why does the UAV remain at rest when the initial velocity is zero?
3. Using the equation of motion, explain why the acceleration is zero.

3 Step 2: Open-Loop Drift

Task

Test:

- $T > mg$
- $T < mg$

Example:

- $T = 1.1 mg$
- $T = 0.9 mg$

Questions

1. What happens when $T > mg$?
2. What happens when $T < mg$?
3. Why is the altitude a quadratic function of time?

4 Step 3: Proportional Control

Task

Use:

$$T = \alpha mg + K_p(z_{\text{ref}} - z) \quad (2)$$

Here, $\alpha \neq 1$ introduces a constant bias in thrust.

Set $\alpha = 1.1$ and $K_i = 0$, $K_d = 0$.

Test two values of K_p .

Questions

1. What happens when K_p is small?
2. What happens when K_p is large?
3. Does the UAV reach the target altitude smoothly?
4. Do oscillations appear?
5. Does the system converge or diverge?

5 Step 4: Derivative Action

Task

Use:

$$T = \alpha mg + K_p(z_{\text{ref}} - z) + K_d(v_{\text{ref}} - v) \quad (3)$$

where $v_{\text{ref}} = 0$.

Set $\alpha = 1.1$.

Test at least two values of K_d .

Questions

1. What changes when derivative action is added?
2. Are oscillations reduced?
3. Does the system become more stable?
4. Does a steady-state error remain?

6 Step 5: Integral Action

Task

Initialize $e_{\text{int}} = 0$.

Use the rectangular approximation:

$$e_{\text{int}} = e_{\text{int}} + e \cdot T_s \quad (4)$$

with:

$$e = z_{\text{ref}} - z \quad (5)$$

Control law:

$$T = \alpha mg + K_p e + K_i e_{\text{int}} + K_d(v_{\text{ref}} - v) \quad (6)$$

Set $\alpha = 1.1$. Test two values of K_i .

Questions

1. What changes when integral action is added?
2. Does the steady-state error decrease?
3. Can the steady-state error vanish?
4. What happens if K_i is too large?

7 Step 6: Conclusion

Write a short conclusion.

Include:

- Open-loop behavior
- Effect of proportional control
- Role of derivative action
- Role of integral action
- Main observations about PID control

8 Submission Requirements

You must:

1. Submit a PDF report using the provided template to KULMS.
2. Push your edited / modified files to your GitHub repository.

The report must include:

- Equations used
- Simulation plots
- Short explanations
- Final conclusion

Follow the template structure exactly. Delete all instruction text in red before submission.